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Presidential Address: The Economist as Designer in the Innovation Process for Socially Impactful Digital Products†

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This paper provides an economic perspective on data-driven innovation in digital products, focusing on the role of complex experiments in measuring and improving social impact. The discussion highlights how tools and insights from economics contribute to each stage of the innovation process. Key contributions include identifying problems, developing theoretical frameworks, translating goals into measurable outcomes, analyzing historical data, and estimating counterfactual outcomes. The paper also surveys recently developed tools designed to address challenges in designing and analyzing data from complex experiments. (*JEL* A11, D83, D91, I20, L80, O31, Q16)

This paper provides an economic perspective on data-driven innovation in digital products. Such products have played an increasingly important role in our lives and in the global economy over the last decades. Moreover, the importance of these products seems likely to increase further in the face of recent innovations in artificial intelligence. In this paper, I argue that digital technology is particularly well suited for addressing a wide range of social impact problems. In addition, economists can play an important role as a "designer" throughout the entire innovation process for digital products. This paper illustrates these ideas through a series of case studies where academics partnered with implementing organizations to design, improve, evaluate, and implement digital products for social impact. Although this paper focuses on social impact applications, most of the issues apply as well to for-profit applications.

As illustrated in Figure 1, I break down the innovation process for digital products into several stages. The process starts with the identification and prioritization of problems and candidate solutions, guided by an economic framework that categorizes the forces relevant for the impact of proposed solutions. It then moves to consideration of how to measure outcomes that relate to social impact goals. Once a solution category is selected, an iterative "data-driven innovation cycle" begins. This cycle involves repeated instances of analysis of historical data to guide design decisions, optimization and refinement of interventions, the design of potentially

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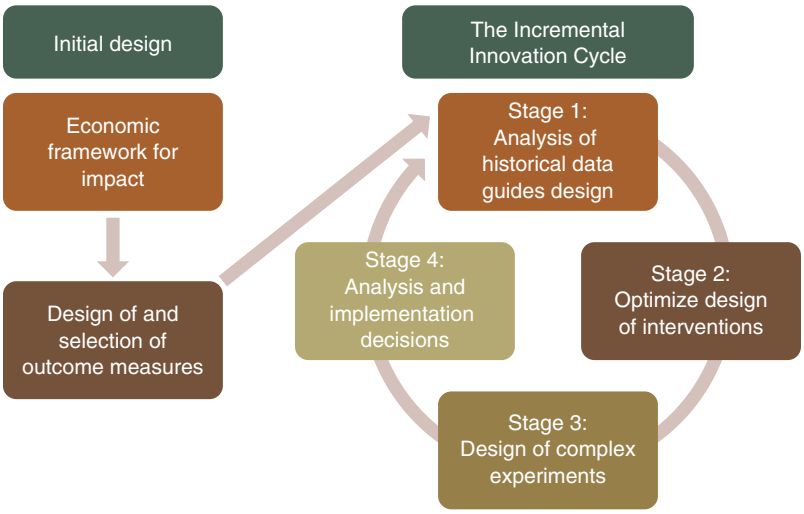

Figure 1. The Data-Driven Incremental Innovation Cycle

complex experiments, and the analysis of experimental data to guide implementation decisions, where implementation may be full or partial (e.g., targeted at specific users). Through examples and case studies, I illustrate how the tools, frameworks, and existing empirical evidence from economics research are used at each stage of the cycle. The specific contributions include the following: problem identification; theoretical frameworks that guide the design of products that address problems and provide a structure for the mechanisms through which impact occurs (including incentives and markets); the design of experiments used to evaluate products; the translation of high-level goals into empirical measures of outcomes; the empirical analysis of historical data; and estimation of the benefits of counterfactual policies.

A central theme in this paper is the role of the economist as a "designer." A designer applies existing ideas and conceptual frameworks and creates new ones when carrying out activities such as identifying problems, as well as breaking problems down into forces or components so as to guide the creation of, and selection among, solutions. A designer generates ideas for solutions and analyzes the strengths and weaknesses of alternatives, iteratively refining the frameworks and ideas for solutions. A designer considers the context where a solution will be deployed. This includes the range of agents, such as institutions and organizations, that will be directly affected as well as those that will be indirectly affected. This also includes the incentives faced by all agents and the new equilibria the solutions may generate. The designer may take actions to shape the context to support proposed solutions.

The economist may play a design role for many distinct categories of problems. In this paper I focus on six design roles and six key cross-cutting challenges facing the economist in these roles, highlighting how economic tools and frameworks help meet the challenges. Of course, the list of roles and challenges is not comprehensive.

The paper proceeds as follows. Section I lays out the design roles and cross-cutting challenges considered in the remainder of the paper. Sections II–VI describe several categories of digital product innovation, each illustrated by several detailed case studies of academic collaborations with implementing partners, previewing issues

that arise in the innovation cycle, the design roles of the economist, and cross-cutting challenges. Section VII describes the innovation cycle in depth, using case studies to illustrate the design roles for the economist and the cross-cutting challenges, as well as some solutions employed in the implementations.

The Supplemental Appendix presents an economic framework for understanding the costs, benefits, and opportunities for investment in innovation in digital products, and in particular it argues that innovation in digital products creates opportunities for both tactical and generalizable research findings.

I. The Economist as a Designer: Opportunities and Challenges

A. *Six Design Roles for the Economist*

In the first design role, the economist serves as a *product designer*, which involves the design of innovations in digital products and services that address social problems. The case study in Section IIIA describes a new digital program created and implemented by the authors that was effective at helping women transition into jobs in information technology (Athey and Palikot 2022). The design of this solution was informed by an economic analysis of the impediments to job transitions. Product design also incorporates some elements of software design, such as decisions about how modular software is and what input parameters (sometimes referred to colloquially as "knobs," "dials," or "levers") can be exposed to analysts inside the firm for optimization and experimentation. For example, the case study in Section VIA describes a setting where PayPal's charitable giving website shows users a list of suggested donation amounts. The lever available for experimentation was the list of donation amounts. Design requires narrowing the set of options to evaluate; in this case, the design decision about which donation amounts to evaluate in an experiment was informed by empirical analysis of historical user behavior as well as by insights from behavioral economics (Athey et al. 2024).

Second, the economist serves as *market designer*, which involves creating and selecting among interventions that shape the functioning of markets or the interaction among individuals in the economy. The role of the economist in market design has been described at length in the economics literature (e.g., Roth 2002, 2018; Milgrom 2021). Section VA describes a case study where Athey et al. (2022a) explore how search algorithm ranking policies affect funding decisions on an online micro-lending platform. The paper analyzes counterfactual ranking policies that incorporate features of profile pictures extracted using computer vision techniques. Because desirable, mutable profile characteristics, like smiling, were correlated with demographics, like gender, changing the search algorithm introduced a market-level efficiency-equity trade-off.

A third role for the economist is *process designer*. This role involves designing and shaping the innovation process, for example, designing decision-making processes and organizational structures that govern decisions about resource allocation, which experiments are conducted, and how decisions are made. The architecture of an experimentation platform also involves many design decisions, including what outcome measures and analyses (such as subgroup analyses) are encoded in software and automatically produced. The architecture of an experimentation platform also

determines what types of experiments are possible (for example, whether the platform supports long-term experiments where a user remains in a treatment arm over repeated interactions with the service). Process design might also involve creating frameworks that guide a mapping between the type of intervention under consideration and the recommended approaches to evaluation of the intervention; for example, some interventions might lead to spillover effects across experimental units, such as interventions for students in a school, which might map into a recommendation that experiments are conducted with randomization at the level of a school rather than at the level of an individual student. Decisions about the design of software architecture interact with this part of the process by determining what types of experiments are feasible to run, as well as what resources (e.g., core software developers versus data scientists) are required to run them.

A fourth role is the economist as a *designer of empirical analyses*, referring to the analysis of historically collected data, either previous experiments or observational data. In the innovation process, the researcher often wishes to prioritize among potential interventions for development and testing. In some cases, historical data can be used to answer questions that inform decision-making. Examples of such questions include: Were similar interventions effective in the past? If so, for which segments of users? Which types of users responded poorly to prior related interventions? The economist can translate these questions into plans for empirical analysis, including plans for how observational data or natural experiments can be used to credibly answer the questions, using tools commonly employed in applied microeconomics research. Sometimes, the intervention of interest is a plan to implement a targeted treatment assignment policy, where there is data available from a prior experiment that randomized assignment to the intervention. Then, the economist can use tools for estimating the benefits of counterfactual policies and estimating optimal targeted treatment assignment policies (see, e.g., Athey and Wager 2021; Zhou, Athey, and Wager 2023). As discussed in more detail in Section VII, the designer may bring together data, models, and empirical estimates from a variety of sources, especially when data is scarce.

A fifth role is the economist as a *designer of outcome measures*, which boils down to operationalizing high-level objectives into specific functions of measured variables. Outcome measures are also referred to in industry and different academic disciplines as evaluation metrics, statistical measures, key performance indicators (KPIs), success metrics, and operational measures, among other terms. The process of defining and selecting outcomes requires a variety of choices, such as which observable variables are aggregated and with what weights, as well as the level of aggregation (over time and across units). The selection process considers a variety of criteria, including both substantive and statistical. Most theory in econometrics and statistics takes the outcome as given. In practice, determining the best outcome measure, or measures, to focus on is a critical design decision. Outcome measures differ in terms of signal-to-noise ratio for evaluating the impact of interventions, thus affecting the required size of experiments and in turn the pace of innovation. Perhaps most importantly, different outcome measures may capture different mechanisms of impact. The literature on field experiments has a history of highlighting the selection of outcome measures as a key decision, as further discussed in Section VIIA. I refer to this activity as the "design of outcome measures."

There are many examples of creative designs of outcome measures in economics, including contingent valuation questions to elicit the value of public goods (Mitchell and Carson 2013), survey questions to learn information about social networks (e.g., Banerjee et al. 2019), and carefully constructed composite outcomes that capture a richer measure of impact than the individual outcomes alone (e.g., Angrist et al. 2020). This paper highlights case studies where survey-based outcomes are used to measure the impact of an educational text message course designed to combat misinformation (Athey et al. 2023b; Offer-Westort, Rosenzweig, and Athey 2024), as well as a study where the impact of a program designed to facilitate labor market transitions is evaluated using observed updates to LinkedIn profiles (Athey and Palikot 2022). Other case studies (e.g., Athey et al. 2022a, 2023c, 2024) grapple with trade-offs among outcomes, where interventions have conflicting effects on different outcomes.

A sixth role is the economist as a *designer of complex experiments*. To start, the allocation of personnel to develop experimentation capabilities, as well as the allocation of experimental subjects to particular experiments, are resource allocation problems. Economists are well suited to analyze the objectives and trade-offs involved in such problems. Experiments can be used to evaluate and select among alternative product design innovations, as well as to guide future innovation, and therefore play a central role in the innovation cycle. Experimental design includes elements such as the structure of the experiment (e.g., does randomization take place at the level of an individual versus groups such as schools or hospitals), the selection of outcome measures, the duration, and the plan for post-experimentation analyses. The characterization of what makes an experiment complex is qualitative, and there are many elements that introduce complexity. One example is the potential for interaction among experimental units when they engage in a network or market ("spillover effects" or "interference"). Section IIA describes a case study where users were exposed to a contest in order to encourage reading. Randomization took place at the level of the school rather than the individual student due to concerns about spillovers (Agrawal et al. 2023). In an example from the for-profit world, Masoero et al. (2024) describes a design for experiments in buyer-seller marketplaces where randomization takes place simultaneously for buyers and sellers. In another example of complexity, Offer-Westort, Rosenzweig, and Athey (2024) designs an adaptive experiment in an attempt to learn the best-performing interventions from a large set of possible interventions that had been proposed in the literature. The adaptive experiment further attempts to learn the best targeted treatment assignment policy, so that individuals are matched to the treatment arm with the highest expected benefit given their characteristics. The current paper describes these and many other sources of complexity in greater detail, highlighting the way in which economic tools, including theory and econometrics, can be used to address design challenges. There are also many open research questions in this area.

B. *Six Cross-Cutting Challenges for the Economist*

Having described six design roles for an economist in the innovation cycle, I now turn to describing six cross-cutting challenges that play an important role in multiple stages of the innovation process and thus influence multiple design problems.

The first challenge is the presence of inescapable, fundamental trade-offs involved in designing and selecting outcome measures, as well as in defining decision rules on the basis of outcome measures. It is rare to find a single outcome measure that can be observed in a timely fashion, has sensitivity to the relevant intervention, has attractive signal-to-noise properties, and yet captures the true social impact of an intervention including all of the mechanisms. Measures that are easily available and timely often miss many channels of impact. In this context, choices of outcome measures have large consequences for the direction and productivity of the entire innovation process. When implementation decisions depend on the outcome measures, the measures influence incentives for those in the organization, and thus the direction and nature of research and development. As Gosnell, List, and Metcalfe (2020) observed, "what is measured is improved." Measures to support decision making thus need to capture concepts that are important to different constituents in the decision-making process, and so multiple measures may be required.

A number of tactics and methods can address these problems. For example, consider the common case where an intervention has conflicting effects on different outcomes. Athey et al. (2022a) and Athey et al. (2023c) analyze trade-offs between efficiency and equity when implementing personalization in recommendation systems. Athey et al. (2024) identifies trade-offs between attracting a larger donor base and raising more money in the short-term when charities raise funds online. All three of these projects suggest constructing a "possibility frontier," similar to a production possibility frontier from classical economics, where different innovation decisions lead to different expected values for two outcomes. Section VIIA provides more detail about how frontiers can be constructed and used. Presenting research findings in this way is itself a design choice; an alternative would be to pick weights and simply analyze the weighted average of outcomes. Presenting a frontier can facilitate organizational decision-making in a setting characterized by trade-offs, and where different decision-makers may have different preferences about how to resolve trade-offs. It can also ensure that product designers pay attention to trade-offs, as design decisions (such as the introduction of personalization) can alleviate them (see Athey et al. 2024).

A second challenge, closely related to the first, is that interventions may directly affect individuals and their outcomes, but the long-term impact of interventions often includes equilibrium effects, as beliefs change and other market participants adapt. Some effects may reinforce and expand positive impacts of interventions, while others may dampen or even reverse positive impacts. It may be difficult or impossible to measure all of these effects in a short-term experiment, and so modeling and heuristics may be used to complement empirical analysis of experimental data. Section VA describes an example of modeling equilibrium effects.

A third challenge arises from the high dimensionality and complexity of the space of possible interventions related to digital products. In the setting of digital products, it is often possible to inexpensively generate a large number of variants of interventions. Then, a limiting factor for innovation is a shortage of users on whom to test variants. This challenge in turn creates opportunities for using models to generalize across interventions, as well as opportunities for complex experimental designs such as adaptive experiments, where poorly performing treatments may be discarded. In one case study, Offer-Westort, Rosenzweig, and Athey (2024) uses an

adaptive experiment and a factorial design to test a large number of interventions designed to reduce the sharing of health misinformation.

A fourth challenge, building from the third, is to unlock the benefits of personalization and targeting of interventions based on individual characteristics. The ability to effectively target strongly increases the returns to designing interventions that may be good for some individuals but neutral or negative for others; and more broadly, it enables each individual to get a product or service that works well for the individual's circumstances. As I argue in more detail below, personalization is complementary to the development of multiple alternative interventions; see also Sverdrup et al. (2023) for a more formal discussion. Making targeting work well with limited data is an additional challenge; in some cases, bringing in external or distinct data sources, or combining results from distinct studies, can help.

A fifth challenge is to manage trade-offs between efficiency and considerations of equity or distributional consequences when delivering digital services in the social impact setting, something that is often salient to decision-makers. Several approaches can address this challenge. First, personalization and targeting can be used to alleviate trade-offs; if an innovation does not benefit a particular group, assignment can be targeted so that the group does not receive it. Second, especially in settings characterized by capacity constraints, methods from causal inference can be used to estimate the effects of counterfactual policies that prioritize particular groups or that seek to improve equity. Sections IIB and VA describe applications of this idea. The tactic of presenting possibility frontiers, described above, can be helpful for improving decision-making processes around trade-offs.

A sixth challenge is the potential tension between innovation that is tactical and thus highly context-specific, and innovation that also creates generalizable knowledge, increasing both the social and the scientific impact of the work. Supplemental Appendix Section III describes considerations that make generalizable knowledge creation more likely to occur for digital innovation. A related issue that has received a great deal of attention in the literature on field experiments is *external validity.* One reason that field experiments may fail to generalize or lead to take-up by implementing organizations is that the experiments are run in an artificial setting, and implementing organizations may not have confidence that the innovations will work in practice (see List 2024 for more discussion). Of course, knowledge must be shared to have general value. Supplemental Appendix Section V discusses the benefits and costs of carrying out innovation in collaborations between academics and implementing organizations, which increases the chance that practical research will be published and shared as well as the chance innovations will be implemented in practice.

Why are economists particularly well suited to meet these design opportunities and challenges? And why do economists ideally play multiple, complementary design roles in the innovation process? Although the answer is multi-faceted, I highlight a few considerations here; for further discussion, see Athey and Luca (2019) who describe the roles economics plays in technology firms.

First, economics as a discipline focuses on combining theory and data to analyze and evaluate business and policy decisions. It is common in economics to start with a framework that incorporates key institutional details; this framework in turn requires that the economist make a determination of what details are important for

incentives and outcomes. The careful crafting of frameworks to match institutions is a hallmark of many fields, including market design, industrial organization, and behavioral economics. In addition, economists are accustomed to analyzing resource allocation problems, and many elements of design can be framed in that way.

Second, and complementary to the first, economists pay particular attention to feedback effects, incentives, and long-run equilibrium, and the discipline has long grappled with the challenges involved in drawing conclusions about long-run effects from limited, observational data. An economist who can iterate back and forth between empirical findings and conceptual economic frameworks can effectively guide choices and decisions, and can prioritize empirical and experimental research that is likely to be most impactful on decisions. Further, an understanding and appreciation of incentives can be helpful in designing organizational processes that generate good decisions. These strengths of economics make them well suited for addressing several of the most important design challenges in the innovation process for digital products, and an economist who can engage in multiple aspects of design can be most effective at anticipating how decisions around one type of design interacts with the others. For example, the selection of outcome measure creates incentives to engage in innovation that is likely to show positive effects according to that measure, even if it might show harm according to other measures. On the other hand, innovations whose benefits are difficult to capture with the chosen outcome measure, for example long-term benefits, might be neglected.

What are some examples of solutions that have been brought to bear so far? In recent years, scholars in economics, statistics, operations research, and adjacent fields have introduced a variety of new methods motivated by the increasing prevalence of and importance of complex experiments in industry. This paper highlights a number of these developments and how they have been applied in practice, where estimation of targeted treatment assignment policies is a central theme (e.g., Athey and Wager 2021). In another example, Athey et al. (forthcoming) develop methods for combining multiple short-term outcomes into a "surrogate index" for long-term outcomes, and they further formalize the assumptions under which estimates of treatment effects based on the index are consistent for long-term outcomes. This paper describes a variety of other approaches. However, there are many open questions remaining in this area, and I expect that it will remain a fertile area for methodological work for some time to come.

II. Application: Educational Interventions

In the area of education and training, digital products have already been widely deployed by both for-profit firms and social impact or nonprofit firms such as Khan Academy. Digital learning can be especially useful for social impact applications due to its convenience and accessibility. For example, adult learners may live or work long distances from in-person learning opportunities. Adult learners may face challenges associated with unpredictable or changing work schedules, or with child care.

A challenge for digital learning is that learners may face difficulty with accountability and motivation, creating an opportunity for innovation to address the challenge. Some popular educational applications employ "gamification" for this

purpose. Section IIA describes an evaluation of an innovation based on this tactic. A challenge for evaluating this type of intervention is determining whether gamification or incentives work in the long-run, or whether they simply boost short-term attention.

Another theme in educational products is personalization. Creating effective personalization algorithms can be challenging for services with a small user base, however. Section IIA describes a case study that overcame this challenge.

A natural application of digital products in the category of educational interventions is in teaching technology and media literacy. The short text message course described in Section IIC educated social media users about misinformation. The design of this solution incorporates many common advantages of a digital intervention: it is cheap to scale, easy to add marginal users, and more accessible for learners than more traditional education interventions.

A. *Case: Personalization and Habit Formation in Educational Technology*

Agrawal et al. (2022) reports on a project where we worked with an educational application in India, Stones2Milestones, with the goal to help children learn to read English in an environment where they might not have access to many books at home. The baseline educational application prioritized stories in a ranked list, curated by humans. The project involved product design, creating the company's first data-driven recommendation engine. Specifically, the product was a targeted treatment assignment policy mapping from historical reading patterns on the app to a ranked list of stories displayed to the user in the app. Here, the economist in the role of product designer also designed the empirical analyses that inform product design choices.

The recommendation engine was evaluated using a randomized controlled experiment that followed children over three weeks, where children remained in the assigned treatment group during the entire time period. Agrawal et al. (2022) report a 40 to 60 percent improvement in the time spent and engagement of children reading stories in the application. Section VIIA below highlights a challenge faced in this project relating to the design of the outcome measure, namely aggregating different measures of "engagement" into a single number.

One concern with the recommendation engine project is that improvements to the application might increase engagement in the short-run, but this short-run increase might simply substitute for long-term engagement. In an extreme example, students may have a limited span of attention for a reading app, and while it may be possible to "pull forward" that engagement, the total time spent might be relatively insensitive to the app quality. In addition, several papers have found null or negative effects on learning when students are provided incentives for learning, suggesting caution when designing approaches for increasing engagement (Lepper, Greene, and Nisbett 1973; Gneezy, Meier, and Rey-Biel 2011).

Agrawal et al. (2023) designed an intervention and a corresponding experiment, with the goal of answering the question of whether increasing engagement in the short-term is enough to create longer lasting habits. The intervention design was inspired by observed practices in larger for-profit applications, where gamification is a popular technique. The intervention involved a contest where students could see

their standing on a leaderboard, and the top 1,000 students at the end of the contest were promised free books.

The design of the outcome was responsive to the concern about short-term versus long-term engagement; the outcome included engagement for 12 weeks after the contest ended. The design also reflected concerns about spillovers or peer effects, which might arise for example if students discussed the leaderboard. Thus, inclusion in the contest was randomized at the school level. In the experiment, schools adopted the app, encouraged usage, and schools in the treated group further alerted students to the contest. The contest was only available to students in the treatment group. In all, approximately 10,000 new students were included in the experiment. Agrawal et al. (2023) found that the students in the contest read more stories during the contest, and most importantly, those habits also persisted for at least several weeks afterwards. In addition, the students read at more advanced reading levels after the contest. These findings contrast with the literature described above that failed to find positive effects from incentivizing learning.

In addition to the direct research findings, this project created tactical know-how and established organizational processes for incremental innovation at Stones2Milestones. Stones2Milestones had not previously gained substantial experience with randomized experiments. The design of the experiments described above included a variety of decisions that can be leveraged for future experiments.

B. *Case: Personalized Call Times for Agricultural Advice*

Athey et al. (2023c) describes a project where we worked with an agricultural advisory service developed and implemented by Precision Development and the Abdul Latif Jameel Poverty Action Lab in partnership with a state government in India. The goal was to design, develop, and implement a recommendation system for the best days and times to call farmers based on their characteristics. In the role of product designer, we wrote and tested the software to estimate a model of the rate at which farmers picked up calls with advice, and we further wrote software to perform the constrained optimization problem to assign farmers to call times while respecting bandwidth constraints for the call system. In the role of designer of complex experiments, we planned a series of iterative experiments, where each experiment tested 91 treatment arms corresponding to a time and day of the week block. The iterative experiments took place over six cycles of approximately 1 week each.

Consider the cycle in more detail. To start, the first experiment randomized the farmers uniformly to the 91 treatment arms. While we had access to historical data, call times prior to the experiment were determined on an ad hoc basis that changed over time, making it difficult to draw valid inferences about which arm was best as a function of covariates.[1]

Next, the data from the uniform random policy was used to estimate an "optimal policy" to maximize user engagement while obeying bandwidth constraints on how many farmers the advisory service could call in a single block. This estimated

[1] Formally, the unconfoundedness assumption likely failed, where the unconfoundedness assumption requires that treatment assignment is as good as random conditional on observed covariates; see, e.g., Imbens and Rubin (2015) for a review.

optimal policy was then tested for the allocation of weekly calls in a second experiment, where one-half of the farmers were allocated to receive call times uniformly at random, and the other half were assigned to the most recently estimated optimal treatment assignment policy. Continuing the cycle, the data from the second experiment was used to estimate a new optimal policy that was then evaluated in the third experiment, and so on for several rounds of iterative experimentation over six weeks.

The data collection strategy enabled two alternative methods to estimating the benefits of personalization relative to assigning call times randomly. Given the structure of the randomization, each of the methods should provide valid estimates of the benefits of personalization, so long as the environment is stable over time. In the first approach, in a given week, the sample mean outcomes can be compared between the treatment group (that received the targeted treatment assignment policy) and the control group (that experienced uniform randomization); this approach can be referred to as "on-policy" evaluation, since the targeted policy of interest was actually implemented. Another approach to estimating the same benefit is to use the purely randomized data from prior weeks to estimate the value of the policy using the tools of "off-policy" (counterfactual) evaluation. Pooling data across weeks had benefits in terms of statistical power, since only a small fraction of farmers in the pure random assignment group received a call at the time that they would have been called under the targeted treatment assignment policy. Athey et al. (2023c) found that the expected benefits estimated using off-policy evaluation tools overstate those from on-policy evaluation, where the latter results are more credible. We explored the cause of this finding, and concluded that preference shocks that change over time can account for part of the discrepancy. We showed that weighting more recent data more heavily better aligns the two approaches. Since instability in the environment is a common feature in practice, this finding may generalize to other settings as well.

While on-policy evaluation has more statistical power and allows for the possibility of testing for shocks that impact the performance of targeted policies out of sample, off-policy evaluation allowed us the flexibility to explore a range of questions, more than we had the ability to test. For example, in this paper, in our role as a designer of empirical analyses, we were able to quantify an equity-efficiency trade-off due to women and men having similar call time preferences, but women being much less likely to pick up a call on average. We found that women's engagement could be increased substantially without hurting the pick up rate for men very much.

In our role as a designer of outcome measures, one concern in the project was that we are measuring a short-run engagement outcome, whether the farmer picks up the call, rather than long-run productivity, the behavior we are trying to impact through calling farmers. Whether picking up the call also improves long-run productivity is not a question addressed in this experiment; other work (on a longer time scale) is required to understand the links between picking up a call and learning the information, between learning the information and implementing the information, and between implementing the information and generating measurable changes in agricultural output. But decision-makers could combine the evidence from this experiment with evidence from another randomized experiments that directly established a positive impact of receiving a call on agricultural productivity (Cole, Joshi, and Schoar 2024; Cole et al. 2025).

C. *Case: Teaching Media Literacy*

Athey et al. (2023b) reports on a project where we collaborated with the nonprofit First Draft News, a coalition aimed at combating misinformation sharing, to develop a five-day text message course that took several different approaches to reducing misinformation sharing. In the role of product designer, we developed the courses to correspond to different stages in a user's decision-making to learn about the mechanisms driving misinformation sharing. The most effective approach taught users to stop and question the content of a post when it evokes a strong emotional response. This approach was more effective than teaching users strategies to identify misinformation, a common tactic that had been studied in the literature, suggesting that intuitive, emotions-driven sharing was driving behavior rather than an inability to identify misinformation. By comparing to a baseline course with facts about misinformation, but no educational content, and to an accuracy nudge shown to be effective in the field, we found that there is value to the skills being taught in the course, as opposed to a simple salience effect or a nudge towards desired behavior.

The project faced several design challenges. One was developing outcome measures; the paper employed a survey exposing users to hypothetical posts, with surveys before and after the intervention to improve statistical power. However, a problem remained that survey outcomes may not correspond to real-world outcomes. To address this issue, the design included for comparison the accuracy nudge, an intervention that had been evaluated to be effective using real-world misinformation sharing rather than surveys as an outcome. A second challenge was assessing the long-term impact of the course; the design included a follow up survey seven to eleven weeks after the original intervention, finding that most of the treatment effect persisted. However, there was a lower response rate in the follow-up survey, increasing the estimated uncertainty about the treatment effects.

III. Application: Demonstrating Individual Skills and Capabilities

There are a variety of ways that digital interventions may affect the process by which a learner demonstrates skills and capabilities. Digital technology may help a learner create resumes or portfolios that demonstrate their capabilities to potential employers. Section IIIA describes a digital program that helped workers create portfolios to demonstrate capability to employers. In another example, Athey and Palikot (2024) created a feature on the platform Coursera that facilitated the sharing of verified credentials from courses.

Digital services may also help job-seekers interact with digital platforms that match workers to employers. In the future, there may be new automated services developed to help workers apply for jobs. On the side of those hiring workers or screening applicants, a variety of services can help screen and test applicants. In the future, it is possible that the nature of applications and screening may be fundamentally altered, with applicant-side algorithms interacting with employer-side algorithms.

A challenge in designing and evaluating interventions in settings such as these is that there is generally limited capacity on the side of those evaluating learners, and so something that improves prospects for one learner may displace another learner.

This challenge also implies that an intervention may be estimated to be effective when evaluated on a small set of learners, but it may become ineffective if exposed to a large number of learners.

More broadly, a credential or portfolio is valuable to the extent that those evaluating it believe that it signals quality. But this implies there may be a minimum scale for a novel credential: to evaluate the credential, employers need to have experience with it or have some other basis for forming beliefs. Furthermore, when the set of learners obtaining a credential changes, beliefs may change as well. Thus, evaluating long-run impact in this area may be a challenging design problem, both at small scale, when beliefs may be based on limited information, but also if large scale adoption is contemplated, since beliefs may adapt and crowd-out may be an issue.[2]

A. *Case: Labor Market Transitions for Women in Technology*

Athey and Palikot (2022) describes a project carried out by academic researchers in collaboration with DareIT, an organization that aims to help women get jobs in the technology industry in Poland. In Poland, the IT outsourcing industry has been growing, but very few women were getting such jobs. The organization had previously implemented a small person-to-person mentoring program, which was limited by the number of mentors. We proposed to them and created a new online program where the women were guided to create portfolios that could demonstrate their expertise.

In the role of product designer, we relied on employer interviews and feedback to address the "chicken-and-egg" problem with respect to how to develop a product that helps women demonstrate competence to employers during the recruiting process when they lack experience. By conducting interviews with hiring managers and human resource experts from prominent information technology firms in Poland, we learned that evidence of practical experience was a critical signal. Thus, we designed the portfolio-based *Challenges* program to serve as a complementary, but more easily scalable, product for the organization to offer alongside the *Mentoring* program. The *Challenges* program involved participants completing six mini-projects to demonstrate job-relevant competencies, curated by domain experts, over 12 weeks in order to create a portfolio.

We carried out two distinct randomized experiments, one experiment to evaluate the *Mentoring* program and the other to evaluate the *Challenges* program. Our outcome of interest was finding a job in the technology industry, which was information the organization had already been collecting through a survey a few months after the program; however, we were interested in longer-term outcomes and knew that survey attrition would impact statistical power. So, in the role of designer of outcome measures, we required participants to share their LinkedIn profile, and then scraped data from LinkedIn monthly to observe employment information for each of the 12 months following the programs. We found that both programs were effective, with

[2] Note that the problem of experimental results failing to generalize is common across many settings. In medical studies, the study population for a clinical trial may be screened and excluded based on other health problems, while when a drug is used in a general population, the health profile of that population may differ, leading to different treatment effects.

the *Mentoring* and *Challenges* programs increasing the rate of finding a tech job from 29 to 42 percent and from 20 to 29 percent, respectively, with no attenuation of the treatment effect over 12 months.

One feature of the program that added a dimension of complexity to the design of the experiment was that mentors choose their mentees, leading to problems of selection. Thus, we implemented a matched-pairs randomization design in which we asked mentors to choose two potential mentees rather than one, and randomized which mentee was assigned to the mentoring group versus the control. The analysis revealed the importance of this choice, as the control group for the *Mentoring* program had better outcomes than the treated group in the *Challenges* program.

Our sample had only a few hundred participants, so in the role of designer of empirical analyses, we relied on the efficiency of a proportional hazards model to obtain our results. Since the timing of finding a job has information, beyond just whether a job has been found at a point in time, the proportional hazards model is able to make use of that information to estimate more precise treatment effects than other model choices. The proportional hazards model, along with the size of the treatment effects, also improved the power of the treatment effect heterogeneity analysis in this small sample, with strong evidence for heterogeneity based on education, professional experience, residence location, and age. For example, we found that those with more than five years of professional experience received a higher than average benefit from the *Mentoring* program, but none from the *Challenges* program, and vice versa for those with less than five years experience.

As in the case study in Section IIB, capacity constraints were important. Both the *Mentoring* and *Challenges* programs were oversubscribed, so one question the study addressed in our role as a designer of empirical analyses was whether targeting is valuable for allocating scarce resources. We addressed the question by combining data from the two experiments, which required applying econometric methods designed for observational studies rather than purely randomized experiments. The answer was affirmative; personalization could increase the probability of finding a tech job by an additional seven percentage points across all applicants.

In the role of market designer, the *Challenges* program is an intervention that improves the information available to employers when matching with workers. One concern when interpreting the results from this project is that we only know the impact of the programs when a few hundred women participate. We do not know what would happen in equilibrium if all applicants had access to one of the programs, or how employers would respond after gaining more experience with a portfolio demonstrating skill like that generated by the *Challenges* program. Interpretation of results can be guided by an economic framework for considering the effects of the program as it scales and as employers gain more experience with it.

IV. Application: Assisting Service Providers

Digital interventions offer a variety of ways to assist service providers, helping to improve both client and provider experiences and client outcomes. For example, digital technology can help service providers and their clients through facilitating the flow, storage, and communication of information, and by helping providers give better and more personalized advice. They can also increase transparency and

accountability, particularly for vulnerable clients, because products used in the course of service create a digital record. The same products can also prevent errors before they happen. Digital products also facilitate tracking progress and synthesizing information. It may thus make counseling more efficient across diverse areas, ranging from health to careers.

A. *Case: A Digital Assistant to Help Nurses Counsel Patients*

Athey et al. (2023a) describes a project where economists and medical researchers from several universities worked with the World Bank and Hôpital Gynéco-Obstétrique et Pédiatrique de Yaoundé (HGOPY) in Cameroon to develop a digital tablet application that would augment the capability of nurses as they counsel patients. This intervention was motivated by the observation that in many developing countries, including Cameroon, there is a shortage of highly trained nurses, and in particular, nurses might not have sufficient knowledge to give the most helpful advice to patients about balancing patient goals with the side effects and inconveniences associated with contraception. Even for an educated nurse, it is particularly challenging to provide advice tailored to the situations and preferences of each individual.

The tablet application was designed to guide nurses through the process of gathering the relevant and most important information about patients and then provide customized information for the patients that was responsive to their situation. By design, a nurse always provided hands-on counseling, supported by the application (and who might learn along the way from his or her experience with the application).

We conducted an experiment with the hospital in Cameroon to increase the take-up of long-acting reversible contraceptives (LARCs) that was cross-randomized on two dimensions of treatment. First, there were two decision-making modes in the application: (i) individual decision-making (IDM) in which the application displayed all available contraceptive options at once in random order, or (ii) shared decision-making (SDM) in which the application displayed the top-ranked method for the patient based on her characteristics, then showed other methods one at a time only if the patient did not want to discuss each prior method shown. The second dimension of randomization was whether and how much the LARC was discounted.

In the role of designer of outcome measures, we were interested in multiple outcomes in this study. First, we wanted to learn which contraceptive method the patient would choose: LARCs, short-acting reversible contraceptives (SARCs), or none. We also wanted to learn how patient experience was impacted, particularly by the personalization in individual- versus shared-decision making process. Furthermore, to understand the mechanism through which the interventions impacted behavior, we wanted to learn about patient search behavior over contraceptive options when faced with individual- versus shared-decision making process. In the long-run, we were ultimately interested in the number of pregnancies, but we did not have enough statistical power to evaluate this outcome. The suite of outcomes provided insight to decision-makers about both patient choices and mechanisms. Measuring patient satisfaction helped alleviate concerns about unintended negative effects of the intervention.

The discounts increased LARC take-up from 11 percent to 29 percent, while SDM more than tripled take-up to 35 percent. This change to services came at no cost to patient satisfaction, which remained at 90 percent for both the IDM and SDM groups. To understand these results, we looked at how many options patients discussed in detail with their providers, finding that SDM patients discussed 0.55 more methods in detail over a mean of 1.15 for IDM clients. These findings are consistent with Athey and Ellison's (2011) theoretical model in which consumers have heterogeneous search costs to meet their individual needs. This model can also explain the striking finding that discounts had no incremental value for patients in the SDM group. Patient exposed to more options in the SDM group may have reduced uncertainty sufficiently in patients on the margin of choosing a LARC to make them less price sensitive.

There was heterogeneity in this price sensitivity, suggesting potential benefits to personalization. While adults aged 20 or older were much more likely than adolescents to adopt a LARC when no discount was offered, offering discounts eliminated this gap. Single women and women who had recently given birth were also particularly responsive to the discounts. In sum, targeting vulnerable populations with discounts is likely to be beneficial.

V. Application: Recommendations in Marketplaces

Recommendations are an integral part of for-profit online marketplaces, as discussed in Section II. They affect the information we find online, the products we buy, and the conversations we have on social media. In the area of social impact, marketplaces and matching platforms (and thus rules and algorithms for ranking and recommendation, ranging from simple to complex) have proliferated. Examples include peer-to-peer lending to disadvantaged borrowers (see Section V), as well as a variety of charitable giving platforms that give directly to recipients (e.g., GiveDirectly). When designing recommendation systems, a number of design opportunities and challenges arise that are relevant to social impact, in particular issues of efficiency-equity trade-offs. The following case study provides an example.

A. *Case: A Peer-to-Peer Lending Marketplace*

Athey et al. (2022a) describes a project designed to analyze potential changes to the Kiva peer-to-peer online lending marketplace, where lenders select borrowers on the basis of online profiles and descriptions. The borrowers have been vetted by affiliates of Kiva and primarily come from developing countries. The research project was motivated by the general observation that online platforms often face challenges balancing efficiency against a desire to avoid disparate outcomes for different groups. Some marketplaces, including Kiva, rely heavily on profile images to promote trust and motivate engagement. However, as artificial intelligence and specifically computer vision is incorporated into recommendation engines and ranking algorithms, there may be unintended consequences. In the case of Kiva, Athey et al. (2022a) documented a positive correlation between certain features of profile images, such as smiles, and funding outcomes. Further, these features were also correlated with demographics, particularly gender, where women tended to create

profiles in "styles" (smiling, lighting, and so on) that are more attractive to lenders. Thus, the paper's results suggest that naively incorporating features of images into profile ranking would increase disparities on the Kiva platform, since women in general are more likely to get funded.

In the role of market designer, we considered interventions to improve both efficiency and equity on the Kiva platform. We were not able to test interventions live on the Kiva platform, and so in the role of designer of empirical analyses we relied on a combination of methods to evaluate hypothetical interventions: the analysis of historical, observational data; a recruited subjects randomized experiment; and simulations from a model calibrated with estimated model parameters. The design of the recruited subjects experiment involved the creation of counterfactual profile images using Generative Adversarial Networks (Goodfellow et al. 2014) that took real images and changed a characteristic of interest, such as whether the person was smiling, to generate a hypothetical profile image. We then conducted a randomized experiment where these features varied randomly across profiles exposed to experimental subjects to estimate the impact of the characteristic on lender demand. We showed that the treatment effects were broadly consistent with estimates of effects of the same features derived from analysis of historical Kiva (nonrandomized) data.

To learn about how search algorithm ranking could improve efficiency and address inequitable outcomes observed both in observational and experimental data, we developed a simple model of interactions on Kiva. Then, we conducted a conjoint analysis on our recruited subjects experiment and estimated parameters of the model. We used this model to generate simulations of the Kiva marketplace and compared alternative interventions from an efficiency-equity perspective. We found that an intervention aimed at nudging *all* borrowers to improve their photos (by, for example, smiling), can shift outward the equity-efficiency possibility frontier.

VI. Application: Nudges in Digital Products

Behavioral nudges are often especially easy to implement in a digital context. They have the potential to encourage beneficial actions for social impact. Nudges in digital products have particular promise in the case of interventions for which there are benefits to personalization. Personalizing small changes in user interface or in the text of a message is feasible in most digital environments.

Nudges are, nevertheless, by their definition small changes. As illustrated in the case study in Section IIC, longer interventions like a multi-day educational text message course can outperform simple nudges when they effectively teach skills. When planning an experiment with behavioral nudges, it is particularly important that minimum detectable effect sizes are compared to effect sizes of similar nudges in the literature, since statistical power can be a challenge.

A. *Case: Personalization in Fundraising Asks*

Athey et al. (2024) describes a project in which researchers were embedded into PayPal's Giving team and ran an experiment on PayPal's charritable giving website. The experiment involved testing two alternatives to the status quo for the list of suggested gift amounts on the "ask string" presented to approximately 400,000

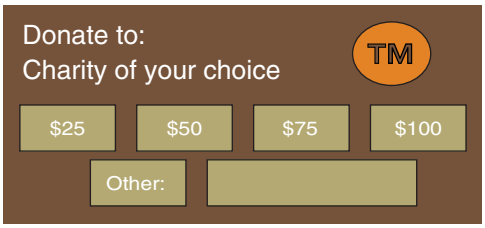

Figure 2. Illustration of an Ask String

PayPal users when they clicked on a charity to donate. These ask strings, like the illustration in Figure 2, are a near universal feature of donation appeals online and in mailings. They are also a low-cost tool for nonprofits to nudge donor behavior. In the role of product designer, we chose the amounts of the suggested gift amounts based on insights from behavioral economics suggesting that different reference points on an ask string might be relevant for different people as well as historical donation patterns.

Athey et al. (2024) studied three outcomes of interest in the role of designer of outcome measures—number of donors (probability of donating), money raised (unconditional donation amount), and user experience (probability of donating a suggested amount)—to learn how the choice of suggested donation amounts can nudge donor behavior. The first two outcomes were motivated by traditional fundraising objectives for nonprofits: raising money and attracting new donors. We also considered user experience because (i) we were interested directly in whether choosing "better" suggested amounts led more users to give a suggested amount, (ii) user experience is an integral part of any digital product (and so should be measured to be improved), and (iii) a better user experience may draw donors back to the service in which they had a positive experience. We found that statistical power for estimating personalized treatment assignment policies was strongest for this outcome.

We found that both of the alternative ask strings increased the probability of donating a suggested amount compared to the baseline ask string (with amounts \$25, \$50, \$75, and \$100) in the 24 hour period following first exposure, but there was a trade-off to changing the highest versus lowest amount. While decreasing the lowest amount (from \$25 to \$10) attracted more donors in the lower reference point ask string compared to baseline, increasing the highest amount (from \$100 to \$200) raised more money in the higher reference point ask string compared to baseline. Both new ask strings increased the subsequent probability of donating and total amount donated in the 4 to 18 months following first exposure.

In the role of designer of empirical analyses, Athey et al. (2024) explored optimal personalized policies for each of the outcomes of interest: number of donors (probability of donating), money raised (unconditional donation amount), and user experience (probability of donating a suggested amount). We identified a trade-off in allocating individuals to the different ask strings: in order to increase the number of donors, the amount raised per donor falls.

However, targeting assignment based on characteristics alleviated the trade-off. By varying the proportion q of the users who were assigned to the higher reference point arm, we estimated 99 alternative personalized policies to map out a

"possibilities frontier" for the two primary outcomes, money raised and donors attracted. Using a personalized policy reduced the amount of money raised that must be given up to gain an additional donor over a random allocation to the lower and higher reference point arms, pushing out the possibilities frontier. The point chosen on the frontier by the nonprofit depends on their relative weighting or prioritization of the two outcomes. See Section VIIA below for more discussion of this type of frontier.

One challenge we encountered when evaluating the trade-off between expanding a nonprofit's donor base and money raised is that immediate treatment effects do not necessarily reflect the long-term benefit of expanding the donor base. By expanding their donor base, a nonprofit can anticipate recurring financial benefits through returning donors and through spillovers created by an extended network of donors. In this study, we examined donation behavior up to 18 months later, but an economic framework for impact might consider the gains to an increase in donor base in terms of a donor's lifetime gift amount. We also do not capture donations made outside of PayPal, so we do not know whether the additional money raised in our experiment was offset by reduced donations elsewhere through a "crowd-out" effect.

B. *Case: Interventions to Combat Health Misinformation*

Offer-Westort, Rosenzweig, and Athey (2024) describes a project that involved running an experiment on Facebook Messenger with 15,000 social media users in Kenya and Nigeria. The goal was to evaluate interventions to reduce health misinformation sharing. In the role of product designer, we aimed to conduct a "horse-race" comparing many different treatment arms to learn about which specific interventions out of the many proposed by researchers and policy-makers worked best, and specifically to learn about interventions at two different levels: (i) interventions that involve platforms labeling posts (headline), for example with warnings or suggestions about related articles, and (ii) educational interventions that can be implemented at an individual level (individual), such as tips about how to spot misinformation. Thus, a treatment arm was defined by a (headline intervention, individual intervention) pair.

To learn about many combinations of interventions, in the role of designer of complex experiments, we divided the study into two phases. The first phase, the "exploration phase," was used to iteratively hone in on the best-performing treatment arm. The experiment in the first phase employed a factorial design with 40 combinations of four headline- and seven individual-level interventions, as well as a headline- and individual-level controls, such that some participants saw both a headline- and individual-level intervention, some saw only one or the other, and a pure control saw neither.

In an effort to optimize power to identify the better performing treatment arms, the experimental design also employed a "contextual bandit" algorithm, which is an adaptive experimental design that learns targeted treatment assignment policies. To be specific, the experiment proceeded in a series of batches, where at the start of each batch, historical data was used to estimate a model mapping individual characteristics to outcome distributions for each treatment arm. The assignment

probability was roughly proportional to the posterior probability that the arm was the best one given the individual's characteristics.[3]

The second phase, the "evaluation phase," conducted a nonadaptive experiment, which collected data about the best-performing individual treatment arms of each type (headline and individual) from the exploration phase, as well as the best targeted treatment assignment policy as estimated at the end of the exploration phase. This approach supported several goals. First, it ensured that the average (over individual characteristics) benefits to the best-performing arm could be precisely estimated. Second, randomizing individuals to those arms created a dataset that would be well-suited for estimation of the value of counterfactual targeted treatment assignment policies. Assigning a portion of individuals to a targeted treatment assignment policy enabled better precision when evaluating that policy, and in particular, ensured that there was enough data for each type of individual interacting with the treatment arm most advantageous for them.

The results from the evaluation phase showed that the average treatment effects for the headline-level interventions were close to zero and were precisely estimated. In contrast, an individual-level intervention that nudged users to consider the accuracy of information effectively reduced misinformation sharing in the survey. In addition to misinformation sharing, we considered other outcomes of interest in this study, varying decisions about aggregation and weighting; Section VIIA provides further discussion. We also explored benefits to personalization between the top two treatment arms and found modest returns to personalizing who was shown the individual accuracy nudge versus the second-best individual-level intervention, which provided a list of educational "tips" about misinformation that had previously been promoted by Facebook.

VII. The Process of Incremental Innovation

This section returns to the innovation process, illustrated in Figure 1. I develop the process in more detail, building on the case studies described above. This section describes the roles of different types of economic approaches, as well as opportunities and challenges for design, at each stage of the process.

As illustrated in Figure 1, the incremental innovation process begins with the Initial Design Phase (Section VIIA), which includes two elements, the Economic Framework for Impact, and the Design and Selection of Outcome Measures. After the initial design phase, an Incremental Innovation Cycle commences (Section VIIB). The innovation process cycles through the four stages repeatedly as the design is incrementally improved. Recently developed econometric methods play a particularly important role in the Incremental Innovation Cycle.

[3] As noted in the literature on policy learning (e.g., Athey and Wager 2021), the quality of a learned policy depends on the extent of "overlap" between the considered set of allowable policies and the collected data. Collecting data using a contextual bandit algorithm can create challenges for policy learning, because as the algorithm stops collecting data about arms that perform poorly for particular types of individuals, overlap can become a problem (see Athey et al. 2022b). Krishnamurthy et al. (2023) develops new algorithms that address this issue in order to improve the quality of policy learning in contextual bandits.

A. *The Initial Design*

An Economic Framework for Impact.—An important initial step in designing a new product or product feature is the development of an economic framework for the innovation's impact. This "theory of impact" is informed by an understanding of the institutional context as well as the history of successes and failures of similar ideas. The context includes the needs, desires, and constraints faced by the potential users of the product, as well as any intermediaries (such as schools, governments, or other institutions) that might affect adoption or usage.

Consider a few of the many ways an economist's perspective can be applied at this stage of product design. First, the designer's understanding of the institutional context can be organized using familiar concepts such as information and incentives, as well as concepts related to economic markets such as the existence and importance of various substitutes and complements for the product. Economic frameworks may suggest consideration of factors like search costs or whether a product is an experience good or is characterized by network effects. Some products seek to improve the functioning of markets, while others seek to influence individual behavior. An understanding of market failures, externalities, and incentives may be important for market interventions, while an understanding of behavioral science may be important for interventions aimed at individual behavior. Section IIIA describes a case where the authors relied on a simple economic framework to come up with an original design for a product to address the challenges faced by women seeking to transition to the information technology sector; employers needed more information about candidates to assess capability for workers who lacked experience, while the workers needed guidance on what investments would overcome this hurdle, and so a program that guided women to create portfolios to demonstrate capability addressed both problems. The specific solution was informed by interviews with potential employers, interviews that uncovered more details about their needs and preferences.

More generally, the economic framework for impact may include the role of various feedback effects or equilibrium ideas. For example, as discussed in Section III, an intervention that involves workers demonstrating their capability with credentials or portfolios may need to address the question of how employers form beliefs about the quality of workers who have the credentials as well as how that might change as a result of scaling the intervention.

The economic framework for impact may also involve changes to equilibrium outcomes in marketplaces. For example, as described in Section VA, Athey et al. (2022a) studies hypothetical interventions in the context of a peer-to-peer lending marketplace for disadvantaged borrowers, mainly in developing countries, on the Kiva platform. In addition to the intervention described in Section VA in which all borrowers would be encouraged to improve their profile image (e.g., through smiling), we considered an intervention that would rank more highly borrowers with profile pictures that are more attractive to lenders. The paper found a potential consequence: since features that make photos attractive are correlated with demographics, this ranking intervention would exacerbate disparities in outcomes by demographics on the platform. An economist might consider the short-term (direct) and long-term (equilibrium) impacts of such a change. Athey et al. (2022a)

identifies an equity-efficiency trade-off and proposes interventions that might alleviate the trade-off and expand the "possibility frontier." This type of framing of the problem, such as identifying a possibility frontier and considering alternative market designs that achieve an objective, is natural when taking an economic perspective on the problem.

As described in Section VA, Athey et al. (2022a) uses a theoretical model calibrated with data from the platform to analyze counterfactual outcomes that would occur in interventions like ranking borrowers higher who have more attractive profile images. However, like most empirical papers, Athey et al. (2022a) does not treat all decisions as endogenous. For example, it takes the participation of different types of borrowers and lenders as fixed, and it does not model changes in lender beliefs that might occur as a result of a systematic change in the behavior of borrowers on the platform when creating profile pictures. Although in practice, an empirical analysis may not incorporate all long-run effects, economic framing can be used to reason about how to interpret more narrow analyses, as well as the expected sign of biases that might arise from holding certain forces fixed. Other settings may involve other types of market equilibrium effects, based on factors such as externalities, dynamics, spillovers, network effects, and so on.

The economic framework for impact also provides insight about categories of intervention that may be helpful. For example, interventions can take place at a high level, such as national regulation that shapes markets and outcomes; at the level of a large firm, which may put in place policies and procedures, approval processes, risk management, organizational design, and incentives; at the level of a marketplace or platform, such as the Kiva peer-to-peer lending platform describe in Section VA; at the level of a firm that has multiple employees, such as a hospital employing many nurses, as in the case study of Section IVA involving an intervention that helps nurses counsel patients; or at the level of an individual or a school, depending on whether the intervention has the potential for spillovers, in a user-facing application such as the educational technology application described in Section IIA.

Design and Selection of Outcome Measures.—The design and selection of outcome measures is a critical step that has important implications for other elements of design, including process design, design of empirical analyses, and the design of complex experiments. More generally, it is a crucial step for any empirical research project. Outcome selection has been discussed in the literature from a variety of angles, including with respect to how the outcome should correspond to the research question (Gertler et al. 2016), the challenges that arise when the indicator of success is less precise or lacks consensus (Glennerster 2017), how to develop indicators specific to the interventions and domains being studied (Glennerster, Walsh, and Diaz-Martin 2018), and the general importance of measuring long-term outcomes (Czibor, Jimenez-Gomez, and List 2019). This topic is also addressed more generally in guidance for conducting experiments (see, e.g., Duflo, Glennerster, and Kremer 2013). The selection of outcome measures is also critical in biomedical applications, where for example there is a literature on surrogate measures, such as the size of a tumor or a measure of cholesterol or blood pressure, in settings where an ultimate outcome such as mortality may be rare (see, e.g., Prentice 1989).

Although at a very high level, social impact organizations may have a single goal, such as "help children learn to read," it is rarely that simple in practice. What does it mean to learn? There may be multiple components of competence, each of which might matter in different circumstances. For example, speed and comprehension may be important in different settings. And organizations may have additional goals, such as instilling a love of reading.

In the context of innovation in pursuit of social impact, one set of trade-offs arises as a result of the fact that the outcome measures that are most aligned with the organization's and society's goals often share several characteristics: (i) they are realized over a long time horizon (for example long-term life outcomes rather than short-term engagement with an application); (ii) they may not be naturally observed through usage of the digital service, but rather they may require additional investment to measure, and it may not be possible to ensure that outcomes can be collected for all users even with investment (for example, labor market outcomes such as wages for participants in digital education); and (iii) the signal-to-noise ratio is low. In such scenarios, the researcher faces trade-offs. For example, there may be an alternative outcome that is quickly and readily observable, addressing issues (i) and (ii), and where the signal-to-noise ratio is high. However, the quickly and readily observable outcome may leave out part of the benefit of the treatment; and in more pathological cases, something that looks good in the short-run might turn out to be bad in the long-run. An example might be a drug with long-term side effects; in the context of digital technology, an application might "teach to the test" but distract time from learning more lasting, fundamental concepts.

If the researcher selects shorter term or more "narrow" outcomes, it may be important to include multiple outcomes in the analysis, where alternatives might capture different effects that would be relevant for predicting long-term outcomes, or where alternative outcomes might be informative about mechanisms. Athey et al. (forthcoming) provide a more detailed analysis of when a set of short-term outcomes, also referred to as "surrogates," can be used as proxies when creating estimates or bounds on long-term outcomes. The methods apply even when the long-term outcome is not observed for the units in the experimental data; rather, the methods enable the analyst to combine an estimate of the relationship between short- and long-term outcomes from an observational dataset, with experimental estimates of the treatment effect of the intervention on short-term outcomes. Specifically, the paper identifies conditions under which short-term measures can be aggregated into a "surrogate index," where treatment effects estimated for the surrogate index are consistent estimates of effects on the long-term outcome of interest. Some of the conditions may be more likely to be satisfied when multiple surrogates are included in order to ensure that long-term effects are captured; however, short-term measures are only valid surrogates if the relationship between short- and long-term outcomes is not affected by the treatment. Since these strong assumptions are required for the surrogate approach to consistently estimate long-term outcomes, Athey et al. (forthcoming) also provides bounds on treatment effects that can be used when these assumptions fail. This approach is just one example where experimental and observational data can be combined to address limitations in evaluating impact using data from small, short-term experiments. Athey, Chetty, and Imbens (2020) provides another approach that can be used in different circumstances, for example in

situations where data from an experiment can be combined with observational data to learn about the relationship between the treatment and both short- and long-term outcomes.[4]

While approaches based on surrogates have promise, they do rely on assumptions. In practice, when it is difficult to capture the long-term outcome in a single surrogate index, the researcher may consider multiple short-term outcomes. For example, multiple short-term measures may be needed to capture effects through multiple mechanisms or channels, all of which might contribute to (or detract from) long-term outcomes, and where the preferences of decision-makers about the importance of different outcomes and mechanisms may vary across decision-makers and over time.

In an educational application like the case study in Section IIA, short-term measures might include time spent, the reading level of stories attempted and completed, and, where available, other measures of student well-being, study habits, and performance across different subject areas. The design of outcome measures interacts closely with other parts of the design process. Economic frameworks can be used to identify the mechanisms and trade-offs, and the design of outcomes may include creative ways to measure proxies for mechanisms through which an intervention improves or harms the ultimate long-term outcome, such as surveys of experimental participants or their service providers (e.g., teachers). Empirical evidence from experiments or analyses of historical data can help shed light on the magnitude of trade-offs, and product designers may also be directly interested in different mechanisms through which an intervention affects different types of users in order to guide implementation decisions and future targeted innovation. For example, if an intervention increases reading scores but decreases enjoyment of students, innovation can be directed at improving the user experience, potentially alleviating the trade-off between the outcomes.

More broadly, there may be trade-offs between different outcomes. For example, in the case study in Section VIA where the goal was to select among alternative suggested donation amounts in charitable fundraising, Athey et al. (2024) found a trade-off between two outcomes, money raised per potential donor and expanding the donor base. When there are trade-offs, policy makers, nonprofit leaders in the case of choosing suggested donation amounts on ask strings, must choose how to balance or weight the outcomes based on their objectives.

Possibility Frontiers: Consider some notation to clarify the exposition. There are N units, indexed by $i = 1, \ldots, N$, where A_i is a treatment and X_i are characteristics. In the simplest case, A_i can take on values in $\{0, 1\}$, but in the case of multiple arms, let $\mathcal{A}$ denote the set of arms. We let 0 be the baseline or control arm. We let $Y_i(a)$ be the potential outcome corresponding to treatment level a. $Y_i(A_i)$ is

[4] Athey et al. (forthcoming) applies to settings where the long-term outcome is not available for the experimental sample and the treatment may not have been utilized in the observational sample. In contrast, Athey, Chetty, and Imbens (2020) applies when there are two datasets, one observational and one experimental, and where both contain information about the same (or similar) treatment and a short-term outcome, but only the observational dataset contains information about the long-term outcome. The methods use estimates from the experimental data about the relationship between the treatment and the short-term outcome to correct for bias in estimates of treatment effects on long-term outcomes in the observational data.

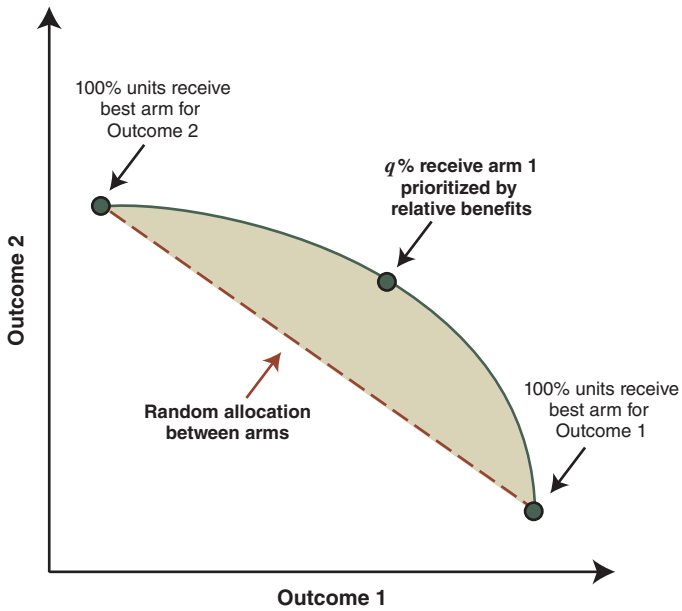

FIGURE 3. HYPOTHETICAL FRONTIER

the observed outcome, sometimes written as Y_i. When there are multiple outcomes, we index them using superscripts, e.g., $Y_i^j(a)$. The Conditional Average Treatment Effect (CATE) of arm a (relative to arm $a = 0$, the baseline) for outcome Y is defined as

$$\tau^Y(x;a) = E\left[Y_i(a) - Y_i(0) \middle| X_i = x\right]. \tag{1}$$

One tactic that is useful when there are trade-offs, as highlighted in several case studies (e.g., Athey et al. 2022a, 2024, 2023c in Sections VA, VIA, and IIB), is the display of a "possibility frontier," which is analogous to a "production possibility frontier" from classical economics. A possibility frontier is a graph where each axis corresponds to one outcome (referred to as "Outcome 1" and "Outcome 2"), so that a pair of values of the form (Outcome 1, Outcome 2) can be represented as a point on the graph. The frontier consists of distinct pairs of values, each corresponding to a different choice about the implementation of interventions. That is, for the case without targeted policies, the frontier is the Pareto frontier of the set $\left\{\left(E\left[Y_i^1(a)\right], E\left[Y_i^2(a)\right]\right) : a \in \mathcal{A}\right\}$. Figure 3 provides an illustration (where the labels in the figure correspond to an example where the intervention involves targeting, the details of which are explained below).

Consider in more detail how such a frontier might arise and be used. As described in Section IA, the operator of a digital application often has access to what are sometimes referred to as "dials," "knobs," or "levers," which are variables or parameters that can be changed at relatively low cost, often by passing values of the parameters as inputs to software. Formally, a knob enables the experimental designer to select from a corresponding set of possible treatment arms $\mathcal{A}$. As part of the Incremental Innovation Cycle (see the discussion of Stage 2 below), the operator can then "turn

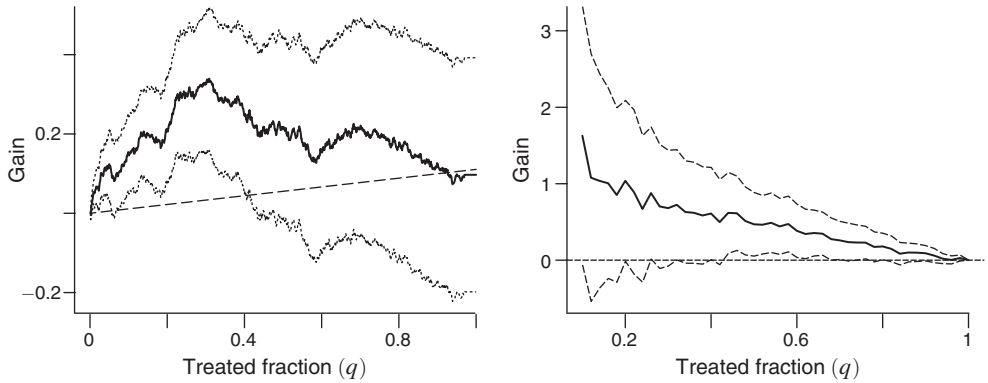

FIGURE 4. QINI (LEFT PANEL) AND TOC (RIGHT PANEL)

the knob" to "tune" the experience in an application by changing the value of these parameters.

When there are trade-offs among outcome variables, a goal of innovation might be to expand the possibility frontier, recognizing that the service may select different points on the frontier in different future scenarios, depending on factors such as the need to attract or retain users. Thus, the goal of innovation might relate to the expansion of the frontier rather than simply achieving a single point on the frontier. The shift from the straight line frontier illustrated in Figure 3 to the curved line is an example of an expansion.

Next, I consider in some detail the concept of a frontier when applied to the case where the intervention is a targeted treatment assignment policy. Formally, such a policy can be written as a function $\pi : \mathcal{X} \rightarrow \mathcal{A}$. In this case, one "tuning parameter" is the share of individuals q assigned to an intervention by a policy. To start, consider an example with only a *single* outcome, and $\mathcal{A} = \{0, 1\}$. Even in the single-outcome case, the concept of a possibility frontier is useful. In this case, the "Targeting Operator Characteristic" curve and the closely related "Qini curve" can be used to understand and summarize the possibility set for the decision-maker as a function of the fraction treated. These are illustrated in Figure 4.

Consider an estimated "ranking function" or prioritization rule $\hat{r}$ mapping characteristics $\mathcal{X}$ to a ranking in $[0, 1]$, normalized so that $\hat{r}(X_i)$ is uniformly distributed. The ranking function induces a policy indexed by the proportion treated q, so that $\pi(x; q) = \mathbf{1}\{\hat{r}(x) > 1 - q\}$. A commonly used ranking function prioritizes units in order of estimates of their CATE, $\tau^Y(x; 1)$ (for $a = 1$). The Qini curve plots, as a function of a budget B for treatment, the difference in expected outcomes between spending B in the most effective way and allocating all units to a costless baseline treatment arm ($a = 0$) (see, e.g., Sverdrup et al. 2023). In the simplest case where costs are linear and normalized to 1, a point on the Qini curve can simply represent the expected benefit from treating q percent of units, ranked by $\hat{r}$. Formally,

$$Q^Y(q; \hat{r}) = E\left[\tau^Y(X_i; 1) \cdot \mathbf{1}\{\hat{r}(X_i) > 1 - q\}\right]. \tag{2}$$

A closely related concept is the Targeting Operator Characteristic (TOC) curve (Yadlowsky et al. 2024). This curve takes a different baseline, namely a baseline where we compare, for a constant fraction q of units treated, the expected outcomes when those q units are randomly assigned to treatment against expected outcomes when they are prioritized by $\hat{r}$

$$TOC^Y(q;\hat{r}) = E\left[\tau^Y(X_i;1)\middle|\hat{r}(X_i) > 1 - q\right] - E\left[Y(1) - Y(0)\right]. \tag{3}$$

The area under the Qini curve represents an aggregate measure of the value of targeting, when the ability to target (or, the ability to target with a particular algorithm or dataset) is evaluated as an innovation. The design decision to prioritize creating Qini curves and presenting them to a decision-maker anticipates that the decision-maker may ultimately carry out cost-benefit analyses of two types: first, when determining the share of individuals q assigned to treatment, and second when determining whether to invest in the capability to personalize treatment assignment.

Yadlowsky et al. (2024) develops a method for measuring the benefits to targeting as either the area under the TOC curve or under the Qini curve, using a measure referred to as the "rank-weighted average treatment effect," or RATE. The use of RATE alleviates the need for the designer to make a choice in advance about how many units will be treated under a targeting policy, and instead produces a single, aggregate measure of the potential benefits to targeting.

Sverdrup et al. (2023) provides methods for estimating the area under the Qini curve for the case with multiple arms (but still a single outcome), proposing a computational algorithm as well as establishing statistical properties for the estimate of the area under the curve. The paper further provides examples to illustrate the extent to which enabling targeting increases the value of using multiple arms, and it gives intuition about how the optimal policy ranks individuals for treatment according to the "bang for the buck" of a particular subgroup of individuals (defined by characteristics) given the estimated costs and benefits of each arm for that subgroup.

Athey et al. (2024) introduces the idea of combining the concepts of a multi-outcome possibility frontier with the concept of a single-outcome area-under-the-curve analysis of the benefits to targeting in the case study described in Section VIA. Figure 3 illustrates a frontier with targeting (the curved line) and without targeting (the straight line). The two endpoints represent $\left(E\left[Y_i^1(1)\right], E\left[Y_i^2(1)\right]\right)$ and $\left(E\left[Y_i^1(0)\right], E\left[Y_i^2(0)\right]\right)$, corresponding to uniform assignment of individuals to $a = 1$ and $a = 0$, respectively. A point on the curve corresponding to q represents the expected values of outcomes 1 and 2 when the policy $\pi(x;q)$ is used: $\left(E\left[Y_i^1\left(\pi(X_i;q)\right)\right], E\left[Y_i^2\left(\pi(X_i;q)\right)\right]\right)$. Athey et al. (2024) carries out such an analysis, where the policy under consideration is a targeted treatment assignment policy allocating units between two treatment arms, and there are trade-offs between two outcomes.

Athey et al. (2024) argues that introducing targeted treatment assignment policies as an intervention, where the fraction treated is a knob, can alleviate trade-offs. To see this, consider a simple example where there are two types of users, old and young. A new user experience in a digital application increases time spent in the application for young users but decreases amount spent for older users. In aggregate, this shows up as a trade-off between time spent and money spent. Targeting the user

experience only at the young users alleviates this trade-off. Without targeting, there are two choices, the old user experience and the new one, leading to two points on the possibility frontier. But targeting creates new points, corresponding to a share treated between zero and one. If the ranking function used to prioritize units for treatments is uninformative, the frontier will be linear, but if targeting is effective, in that the high-priority individuals have higher treatment effects, the possibility set will expand outward.

Formally, consider the incremental value of adding another arm, a', to a set $\mathcal{A}$. If the researcher only considers uniform assignment policies, where all individuals receive the same treatment arm, then arm a' has no incremental value unless $E\left[Y_i(a')\right] > \max_{a'' \in \mathcal{A}} E\left[Y_i(a'')\right]$. On the other hand, with targeting, a' has some value as long as there exists a subset $\mathcal{S}$ of $\mathcal{X}$ where $E\left[Y_i(a') \middle| X_i \in \mathcal{S}\right] > \max_{a'' \in \mathcal{A}} E\left[Y_i(a'') \middle| X_i \in \mathcal{S}\right]$. See Sverdrup et al. (2023) for more discussion.

Athey et al. (2024) utilizes these types of frontiers when evaluating the benefits of targeting the suggested donation amounts exposed to donors in charitable giving. We show that targeting can alleviate the trade-off (that is, shift out the possibility frontier) between expanding the donor base and raising more money in the short-run, with average annual donation amount driving the heterogeneity, along with correlated demographics like age and income.

As outlined in Athey et al. (2024), a decision-maker ultimately needs to select a value of q corresponding to a point on the frontier, which is equivalent to selecting weights between two outcomes. This follows since (ignoring costs) at an optimal choice, the slope of the frontier is proportional to the ratio of weights. However, many nonspecialist decision-makers have difficulty precisely quantifying the weights on different outcomes in advance of making a decision. Thus, when considering the decision process of an organization, producing a frontier can facilitate narrowing in on the relevant choices for a decision-maker. It can also help decision-makers translate weights on objectives into decisions and their implications.

Aggregating Outcome Measures: In several case studies, we created an aggregate measure composed of a weighted average of several measured outcomes, or one that balanced a beneficial outcome and a harmful outcome. Agrawal et al. (2022) created an overall measure of student engagement with stories using a weighted average of measures such as clicks and story completions, using weights suggested by the application developer Stones2Milestones. Offer-Westort, Rosenzweig, and Athey (2024) and Athey et al. (2023b) consider two outcomes, intent to share misinformation and intent to share non-misinformation. The primary outcome in each case, referred to as "discernment," depends positively on sharing non-misinformation and negatively on sharing misinformation, but in general the evaluated interventions tend to reduce both, leading to a trade-off and potentially making results sensitive to the weights. The papers show both aggregated and disaggregated measures to address this concern. Offer-Westort, Rosenzweig, and Athey (2024) also shows that it is easier to estimate targeted treatment assignment policies that show statistically significant benefits from targeting when the policy is estimated by optimizing for the outcome of misinformation sharing; attempting to estimate policies that optimize treatment effects on discernment produced weaker results, due to low signal-to-noise ratio in the data. Athey et al. (2024) highlights a similar theme: the signal-to-noise ratio for

estimating targeted treatment assignment policies is better for some outcomes than others. Even when the goal is targeting to optimize for one outcome, it can sometimes be better (from the perspective of that outcome) to estimate a policy with the objective of optimizing treatment assignment with respect to an alternative, less noisy outcome.[5]

Aggregating over Units and Time: Another issue in the design of outcomes, one that is also central to experimental design, is the level of aggregation of the units where we measure the outcome. Frameworks from economic theory can be used to organize institutional facts to guide the choice of what experimental unit a pilot or experimental evaluation should focus on. If there are network effects or equilibrium effects, ideally an experimental unit would incorporate these effects, so the best level of aggregation might be a distinct local geographic market or subnetwork that is fairly self-contained. Examples include cities, countries, hospitals, schools, or collections of products, services, or jobs that compete with one another.

However, aggregating units can lead to concerns with statistical power. Sometimes there just are not enough units at a higher level of aggregation, and the analyst may choose to experiment at a lower level of analysis, like an individual customer, and rely on modeling or heuristics to account for long-term equilibrium effects. Even absent corrections for spillovers, short-term or partial equilibrium effects may place a lower bound on effects when equilibrium effects are expected to be beneficial. On the other hand, if short-term and long-term effects might go in the opposite direction, it becomes more important to gather empirical evidence about the full set of equilibrium effects.

A second question about the level of aggregation concerns time aggregation. In many settings where a digital technology is evaluated, a user may be observed using an application many times after a treatment. One alternative, as implemented by Agrawal et al. (2022) in the case study in Section IIA, is to prespecify a time period in which to observe the individuals and select as the outcome the user-level aggregate of the outcome over the time period. Another option is to aggregate the data into smaller, fixed time intervals, such as days or weeks, and plan to analyze the data as a panel dataset, with time-varying treatment effects that may incorporate the decay or accumulation of treatment effects. For example, it may take time for a user to learn how to use a new feature in an application; but a user may also show initial curiosity about a new feature, and then decrease engagement later. With sufficient data, the time structure of treatment effects can be estimated without any additional assumptions, but with limited data, functional form assumptions may be helpful. For example, assuming a linear decay of treatment effects or using a proportional hazard model, as in the case study in Section IIIA, to model rare binary outcomes may reduce the number of parameters estimated. The designer also needs to consider how long to continue collecting data, where trade-offs arise regarding attrition as well as the decay of effects. Although some of these decisions can be deferred

[5] The theme is also explored by Athey, Keleher, and Spiess (2025), which shows that targeting based on simple functions of predicted outcomes and estimated treatment effects can perform better than estimating policies based on flexible estimates of treatment effects for the outcome of interest, because the latter approach suffers from a low signal-to-noise ratio.

until after the data has arrived, the analysis strategy affects the design choices for the experiment, including its sample size but also other elements of design. For example, if the plan is to estimate how the treatment effects decay over time, there may be value to gathering data for a longer time period, while if the plan is to focus on, say, the aggregate outcomes in the first two weeks, lengthening the experiment may only be useful for supplementary analysis. Overall, there is a trade-off between efficiency and statistical power that can be gained from functional form assumptions versus complexity and the potential for bias introduced by these assumptions.

More statistical power may be gained by dis-aggregating the data and analyzing it as a panel in several circumstances. First, it is common that units commence treatment in batches, or as they arrive to use an application, for applications that users make use of irregularly. Thus, the data takes the form of a "staggered rollout," something that has received much attention in the literature on causal panel data models (e.g., Athey et al. 2021a; Xiong et al. 2024). In such settings, modeling underlying time trends and unit characteristics can improve statistical power. This statistical power may come at the cost of additional modeling assumptions, particularly functional form assumptions such as additivity of time effects. Unlike the observational setting, random assignment of the treatment implies that it is not necessary to estimate unit fixed effects or latent factors in order to analyze the impact of the treatment. In settings where there is substantial variation in outcomes that is a predictable function of time, and where there is a modest number of units, the panel data setup may be more attractive. Xiong et al. (2024) provides a formal analysis of how to optimize the design of a staggered rollout experiment.

B. *The Incremental Innovation Cycle Stage by Stage*

After the initial design phase, innovation shifts into a cycle of incremental innovation. This cycle starts from the point where there is already a digital product or service that has been created in some baseline form, and where an implementing organization has the ability to collect data and run experiments to evaluate changes to the product. Some of the stages described here can be applied even without those conditions, however.

In addition to frameworks from economic theory and knowledge gained from prior empirical studies, frameworks and methods from econometrics and statistics play a central role in the design problems that arise in the steps of the Incremental Innovation Cycle. Methods for estimating causal effects are particularly important, where recently a number of tools have been developed that are motivated by problems encountered in digital experimentation. Some of these tools have already been widely deployed in both academia and industry, including in many of the case studies described in this paper.

For any particular iteration of the Incremental Innovation Cycle, the first design issues require revisiting the steps for initial design from Section VIIA, but now focused on the specific improvements under consideration. Thus, a first question concerns the economic framework for how the incremental improvement will impact users. Is the improvement designed primarily to have a short-term, immediate effect, for example making some information more visible to the user or surfacing particular information? Or is the improvement designed to encourage the

user to engage more deeply in the application or to gain some knowledge that will help them improve a longer-term outcome? The answers to these questions guide the design decisions for the outcome measure used in the Incremental Innovation Cycle, for example, determining its level of aggregation (e.g., a user, a market, or a time period).

Stage 1: Analysis of Historical Data Guides Design.—The first stage of the Incremental Innovation Cycle uses historical data to generate ideas for improvements to a product, and to evaluate and prioritize such ideas. Historical data may include purely observational data, or data from prior experiments, but in general the improvements under consideration at this stage may not have been attempted before for real users. A new feature might be an improved personalized recommendation system, as in the educational application case study of Section IIA or the case of introducing personalized call times for agricultural advice in Section IIB, or it might be a change in information exposed to a user, as in the case study in Section VIA where different suggested donation amounts are exposed to users.

The economist plays a role in the design of empirical analyses at this stage. Given the design of the outcome measure, the design may specify that descriptive statistics and predictive models are used to describe characteristics of users and help prioritize segments of users where there may be the greatest potential for improvement. For example, the empirical design might specify the use of historical data to predict which users are most likely to abandon the application. The analysis might divide the existing user base into five quintiles according to likelihood of abandoning, and then describe the characteristics of those most likely to abandon, searching for patterns that might suggest possibilities for improvement. For example, if users for whom English is a second language are likely to abandon, that suggests that further analysis could be carried out to assess whether improvements to the user interface would help. Another way such data can be used is to identify clusters or groups of users that have similar characteristics. For example, a "k-means clustering" algorithm based on characteristics can be used to create segments of users. The product design process can be more effective when it is specialized for a particular segment of users that share similar preferences and behaviors; the product designer can consider each segment of user, one by one, to ensure that there are at least some groups for whom the product provides the desired benefits and potentially design multiple products to fit different segments.

Another way to identify opportunities for improvement is to use historical data about previous introductions of improvements that may have similar characteristics or hypothesized benefits. This data may come from one of a number of settings such as: randomized experiments, possibly stratified; a "staggered rollout," where an innovation is made available to different groups of users at different times, eventually becoming available to all users; a simultaneous rollout, where all users are exposed to a change at the same time; or an adaptive experiment. The design of the empirical analysis here would leverage tools from the econometric literature on causal inference in observational studies (see, e.g., Imbens and Rubin 2015; Wager 2025).

Consider an example where historical data from a (simple) randomized experiment is available, where users were randomized into a treatment group and a control

group. Then, the design of the empirical analysis may specify the use of data-driven techniques to estimate and draw insight about treatment effect heterogeneity (Athey and Imbens 2015; Athey, Tibshirani, and Wager 2019; Wager and Athey 2018).

The goal of this analysis would be to answer questions such as: As a function of pretreatment characteristics (covariates that are exogenous to the treatment), how do estimated treatment effects $\hat{\tau}(x; a)$ vary with x (following the notation from Section VIIA)? Understanding for whom an intervention works and for whom it does not can provide insight that informs future innovation.

Another type of innovation that may be considered is the introduction of a targeted treatment assignment policy, $\pi : \mathcal{X} \rightarrow \mathcal{A}$. Methods for estimating policies $\hat{\pi}$ have recently been the subject of substantial methodological research (Manski 2004; Kitagawa and Tetenov 2018; Zhan et al. 2024; Athey and Wager 2021; Sverdrup et al. 2020, 2023; Krishnamurthy et al. 2023).

Some examples of hypotheses that may be tested that relate to targeting and personalization include the following: (i) Does the best estimated targeted policy $\hat{\pi}$ improve over the control (baseline) ($E\left[Y_i\left(\hat{\pi}(X_i)\right)\right] > E\left[Y_i(0)\right]$)? (ii) Does the best estimated targeted policy improve over simply allocating all units to the treatment arm that performs best on average ($E\left[Y_i\left(\hat{\pi}(X_i)\right)\right] > \max_{a' \in \mathcal{A}} E\left[Y_i(a')\right]$), that is, is there a value to personalization? (iii) What are the confidence intervals for the estimate of the value of the best estimated targeted policy? (iv) Which characteristics differ across groups assigned to different arms by an estimated assignment policy (e.g., what are the average characteristics of units in the set $\left\{x : \hat{\pi}(x) = a\right\}$? (v) What characteristics differ across groups that have low or high treatment effects? (vi) If the average treatment effect for an arm is positive for one outcome and negative for another, can targeted treatment assignment alleviate the trade-off between the two?

To operationalize and provide more detail about questions (i) and (ii), researchers may make use of methods described above in Section VIIA, such as creating TOC and Qini curves, and calculating the outcome measure RATE in order to quantify the benefits to targeting.

One challenge for this type of exercise early in the life cycle of a product is that limited data is available. Recently developed approaches from artificial intelligence can sometimes be helpful. In particular, foundation models (Bommasani et al. 2021) can be estimated using large, unstructured, and unrelated datasets, enabling them to learn lower-dimensional "representations" of high-dimensional data. The most popular foundation models today involve text, and they represent sequences of words (which for N words could involve about $30,000^N$ distinct potential sequences) by real-valued vectors with a few thousand elements. Vectors that are nearby in Euclidean space have similar meaning. As reviewed in Du et al. (2024), economists have already found a variety of applications of foundation models. For example, a small dataset might include text reviews or feedback from customers. That dataset may be too small to be used to classify whether reviews are favorable just based on keywords. But, when a foundation model is applied or possibly "fine-tuned" on the data, the analyst may be able to build a useful classifier.

As Vafa, Athey, and Blei (2024) show, these techniques can be used not just for predictive models, but also for causal inference and estimation of heterogeneous treatment effects. Future work may further expand on the toolkit available

to researchers who wish to combine data from small experiments with foundation models or other widely available observational data.[6]

Stage 2: Optimize Design of Intervention.—The first stage in the Incremental Innovation Cycle results in a decision about what interventions should be prioritized for further refinement and development. The second stage involves additional investment in product design. This stage develops and optimizes the intervention, informed by theory, prototyping, additional data gathering, and analysis. It may also involve the development and testing of new algorithms and software.

One type of optimization that often takes place at this stage of the incremental innovation process is designed to inform the "tuning" of the "dials" discussed in Section VIIA. Historical data can be used to estimate the counterfactual benefit from many different values of tuning parameters, and this analysis can be used to prioritize randomized experiments which can produce more credible estimates with more statistical power to fine-tune the design choices. If the design of the software architecture supports passing different values of tuning parameters, testing alternatives can be carried out without large investments of time and development resources. Thus, the design and architecture of the software that implements digital products interacts strongly with the productivity of this type of incremental innovation.

For interventions that might involve more substantial (and costly) software development to implement, a common practice before investing in production-quality development is to use alternative forms of data gathering that do not require engineering effort. For example, recruited subjects or focus groups may be shown screenshots from a hypothetical product or feature before it is actually built. They may be asked to make choices between alternative versions of products, or to express their willingness to pay. In the case study described in Section VA, subjects were recruited via an online survey platform and presented with alternative profiles of borrowers, then asked to choose which borrower they would select for a loan. This approach is substantially cheaper than running a live experiment on users, although of course the external validity of the survey may be questionable. In another example, when developing the text message course described in the case study in Section IIC, pilot experiments using recruited subjects were used to refine the intervention.

When the innovation under consideration is an algorithm, such as a recommendation system or a targeted treatment assignment policy, then this stage of the Incremental Innovation Cycle typically involves estimating or "training" statistical models and using them as inputs into targeted policies or recommendation algorithms. At this stage, methods for evaluating counterfactual targeted treatment assignment policies can be used to compare alternative approaches.

A challenge arising in the case study from Section IIA, which evaluated the introduction of a new recommendation engine for stories, is that there are limits to what can be accomplished with counterfactual evaluation due to lack of "overlap" between the counterfactual policy under consideration and the historical data that was collected. For example, if certain stories were only shown to heavy users in the past, because they were consistently ranked low in the list of stories exposed to

[6]There is a long history in causal inference of combining data sources, for example this is done in the split sample instrumental variables method (Angrist and Krueger 1995).

users, then it may be difficult to estimate how light users would respond if shown those stories. In addition, if a new algorithm is introduced in the application, then users will be exposed to a sequence of experiences with stories assigned by the new algorithm. This might affect overall engagement with the app. It is thus challenging to use historical data to estimate the impact on longer-term engagement from prolonged exposure to a new algorithm. As discussed in Agrawal et al. (2022), this issue motivates designing and running experiments where users do have sustained exposure to a new algorithm. Such experiments require that treatment assignment in the experiment is assigned at the user level, rather than at the level of a user session in the application.

Stage 3: Design of Complex Experiments.—The second stage of the Incremental Innovation Cycle results in a set of interventions to be tested. Stage 3 takes on the problem of designing a (potentially complex) randomized experiment to evaluate the benefits of the interventions.

An initial step in this stage is to consider the goals of the experiment. Most experiments have as their primary goal the evaluation of the impact of an intervention on a set of specified outcome measures, and that will be our focus here. However, experiments may have secondary goals as well. If the analyst looks ahead to future iterations of the Incremental Innovation Cycle, the question of how the data will be used in the future to evaluate new interventions comes into play; for example, the experimental design may anticipate conducting several of the Stage 1 analyses described above. The reuse of the data for future estimation and learning, both for the interventions under consideration in this iteration of the cycle and for innovation in future iterations, affects the design. For example, it creates additional value to including treatment arms in the experiment whose performance may provide insight about future product development, and to ensuring that there is sufficient statistical power for future analyses that may reuse the experimental data.[7]

The pre-experiment planning process goes far beyond just specifying hypotheses to be tested (and, with an eye towards creating generalizable scientific knowledge, potentially preregistering these at a public repository such as the AEA Randomized Control Trial Registry.[8] Planning can be extensive, ensuring that the learnings from the experiment will be worth the cost, and it may anticipate multiple goals for the experiment that may be operationalized by measuring multiple outcome measures, as well as by carrying out more complex analyses such as those described in Stage 1 of the Incremental Innovation Cycle.

One key part of the experimental design process is to select among alternative outcome measures and validate outcome measures in various ways in the context of the proposed experimental design. As described in Section VIIA, there are several important considerations and trade-offs in selecting the outcome that interact with

[7] Note that algorithms for adaptive experimentation, such as bandits or contextual bandits, are explicitly designed to factor in the benefits of "exploration" when selecting treatment assignment policies at a point in time. One perspective is that a bandit is an automated version of the Incremental Innovation Cycle, where the data collection, analysis, and determination of treatment assignment probabilities is carried out by the algorithm. See Athey et al. (2022b); Offer-Westort, Rosenzweig, and Athey (2024); Xiong et al. (2024) for more examples and references for adaptive experiments.

[8] See https://www.socialscienceregistry.org/.

the design of an experiment. The researcher may have a hypothesis about plausible treatment effect magnitudes, or how large they need to be to justify a subsequent investment in implementing the product feature outside the experimental context. These considerations, together with the size of the user base available for the experiment, may rule out focusing primarily on outcomes that are measured with too much noise or that are not hypothesized to be sufficiently sensitive to the intervention, or where there may be insufficient power to carry out supplementary analyses.

A particularly important post-experiment analysis in the digital context is motivated by the fact that it is often possible to deploy the intervention selectively, only to users for whom it is found to be beneficial. Planning an experiment with sufficient power and the right data to estimate a targeted treatment assignment policy is more complex, and generally experiments must have larger sample sizes. To see why, note that even if there are only two types of users, say young and old, determining that one treatment arm is better for the young and another is better for the old requires estimating the treatment effects for each group, and then estimating the difference between those treatment effects. An area where future research is needed is in designing such experiments. Common practice today is to assess statistical power using simulations, or to focus on power for some simple hypothesis tests, such as tests for differences in treatment effects or differences in estimated optimal policy assignments across prespecified subgroups. The latter approach is helpful because typically in cases where targeting is beneficial using more complex, data-driven policies, targeting is also beneficial when policies are defined with simple subgroups. The problem is that it is difficult to know in advance what subgroup definitions yield the greatest benefit to targeting. However, if an experiment does not have sufficient statistical power to detect economically meaningful differences in policy assignment across a given prespecified subgroup, there may also be challenges in estimating more complex policies. More automated and expansive methods that more directly anticipate the goals of estimating targeted policies (and perhaps multiple goals) for post-experiment analysis would be useful for practitioners, and this represents an opportunity for future methodological research.[9]

A number of other considerations for complex design arise depending on the specifics of the setting. I review some of them below.

Multiple Arms: A challenge that often arises when designing or improving digital products is the large number of potential interventions. This challenge can arise, for example, when a digital intervention is composed of multiple components, and there are many options for each component. For example, the user interface in an application often has multiple modules; and an email or a text message course can vary in style, tone, theme, the use of images, the behavioral tactics employed, the subject line, and so on. Of course there are many ways to make a physical product as

[9] As described in footnote 7, one perspective on contextual bandits is that they automate the design of experiments to learn targeted treatment assignment policies. See Krishnamurthy et al. (2023) for a recent algorithm designed to learn such policies. However, these bandits do not directly allow the analyst to specify multiple hypothesis tests of interest, something that Athey et al. (2022b) does via an extensive pre-experiment simulation study using pilot data to guide the simulations.

well, but in practice it may be prohibitively costly to build many prototypes. In contrast, with digital interventions it may be straightforward to generate a large number of interventions characterized by different choices for each component. Then, the limiting factor becomes the amount of data available to learn about the benefits of each intervention, and in particular the number of users of a product available for experimentation.

Several approaches have been employed to address these challenges. First, researchers might place functional form assumptions (such as additivity) on how outcomes depend on characteristics of products, and then model expected outcomes as a function of these characteristics using data from experiments where the characteristics are randomized independently. These "factorial" experiments can allow researchers to explore a wide range of treatments. If all dimensions are randomized independently, it is possible to analyze the effect of changing one dimension at a time while treating variation in other dimensions as random background noise. However, the factorial design has costs because the variance of the estimate of the treatment effect in one dimension is higher when other dimensions may vary. If the researcher is willing to impose functional form assumptions, such as additivity of outcomes in different dimensions of intervention, efficiency gains are possible. Although the functional form assumptions are strong, they are also testable, though statistical power for testing these assumptions may be limited, since there may only be a few observations for each combination of components. See the case study from Section VIB for an example of a factorial design in practice; Offer-Westort, Rosenzweig, and Athey (2024) provides additional references.

Another approach, discussed next, is to use adaptive experiments to narrow down the set of alternatives.

Adaptive Experiments: As discussed in Section VIIA, another approach to the challenge of multiple arms is an adaptive experiment, or a "bandit." Such experiments can be helpful by, for example, discarding poorly performing arms, and thus allocating more individuals to better performing arms. Bandits may or may not be "contextual" (that is, they may or may not involve targeting treatments according to individual characteristics). See Athey et al. (2021b) for a practitioner's guide to simple (noncontextual) bandits. Section VIB describes an adaptive experiment to test a factorial design in practice. In another implementation, Athey et al. (2022b) assesses the ability of contextual bandits to improve over traditional randomized controlled experiments at learning treatment assignment policies, finding that traditional algorithms did *not* lead to improvement, and that a modified algorithm gave only modest improvement. Krishnamurthy et al. (2023) develops an algorithm that, guided by theory, shows greater promise for policy learning.

Adaptive experiments are closely related to problems where an individual is treated over time, for example, in a medical setting where a patient's treatment responds to the evolution of symptoms. Treatments are assigned over time as a function of past treatments and outcomes. For example, chatbots that dynamically respond to a patient's health indicators have been shown to help users lose weight or provide timely information relevant to a patient's health condition (Chang, Shih, and Kuo 2022). Educational platforms also employ dynamic, state-contingent treatments based on a student's progress on a specific concept or subject area.

Marketplace Experiments: As discussed above in Section III and VIIA, the effect of implementing an intervention "market-wide," for example, for all workers in a labor market or all participants in an online marketplace or a social network, differ from the sum of effects of the intervention on individuals in a context where other individuals are not treated or have not been treated very long. One example that has received attention in the literature is the example of a social network; an intervention may lead an individual user to engage less in a social network holding fixed the behavior of the user's peers, but if all of the individual's connections also receive the intervention and engage less, the user may not find as much useful content, leading to an even greater drop in engagement.[10]

Capturing all long-run equilibrium effects in a market or network setting is frequently too difficult for an experiment; thus, a common practice is to combine complex experimental designs that capture some of the forces and enable measurement of key parameters with modeling and/or heuristics that translate the estimated parameters into estimates of longer run effects. These approaches can take the form of estimated structural models, as is common in the field of industrial organization (Hortaçsu and Joo 2023) or in calibrated models in macroeconomics.

There are a variety of other tactics that can be combined with these approaches. As described above, alternative sources of measurement like surveys can capture beliefs and intentions of participants. Such research, combined with analysis of historical data, can be used to create behavioral population segments that can help forecast heterogeneous responses of individuals. Tools from the literature on heterogeneous treatment effect estimation can similarly help create more refined estimates that can be used as an input to modeling or heuristics.

Another approach is to aggregate the data and run experiments with randomization at the level of a unit that might incorporate most spillovers, for example a city or a school (Crépon et al. 2013). Given that this often limits the number of units and creates problems for statistical power, and that units may have time trends that vary by unit, staggered rollout experiments (Hemming et al. 2015; Xiong et al. 2024) may be a useful tactic. A carefully optimized rollout design enables the analyst to estimate both time effects and unit effects, increasing power. The design can also be customized for the case where treatment effects may grow over time, as when there is a delay while a market adjusts to a new equilibrium following an intervention. See Xiong et al. (2024), who develops an algorithm for designing such experiments.

A recent literature has developed to tackle the design of experiments in marketplaces more directly; see, for example, Masoero et al. (2024); Munro, Wager, and Xu (2021) and references therein.

Stage 4: Analysis and Implementation Decisions.—Once data from the experiment has been collected, it can be analyzed to test the hypotheses that were specified in Stage 3. The most natural analyses will compare a treatment and control group for the various outcomes selected in Stage 3. As described above, in some cases, there may be trade-offs between different outcomes, in which case the analyst must

[10] Experiments with interference have been a popular research topic in recent years, see, e.g., Crépon et al. (2013); Athey, Eckles, and Imbens (2018).

choose whether to display possibility frontiers and highlight trade-offs, or to focus on a weighted average of outcomes.

A design problem for this stage arises for the analysis of the experimental data in order to inform decisions about whether, and for whom, the intervention will be implemented. If the software architecture of the digital product supports targeting interventions based on characteristics, the decision-maker does not need to simply decide whether or not the best-performing intervention (or the baseline control) is implemented for all users; they can instead decide to implement a targeted treatment assignment policy, whereby each user is assigned to an intervention based on characteristics according to an estimated treatment assignment policy. The analyses described in the discussion of Stage 1 can be used to estimate a policy and guide this decision.

Whether or not targeting is possible, it still may be useful to decision-makers to understand which users are helped and which, if any, are harmed by an intervention. In social impact applications, distributional effects may play a particularly important role. Thus, the analysis may include an exploration of treatment effect heterogeneity using the tools described in the discussion of Stage 1.

Another important question for the analysis of experimental data is how to advise decision-makers when important forces identified in the economic framework cannot be measured using the experiment. One example is a long-term effect where there is no short-term proxy; another is an equilibrium effect in a setting where it is difficult to intervene at the market level (or where the product serves only one market). The case study about the Kiva peer-to-peer lending platform described in Section VA combines experimental and observational estimates of lender demand for borrowers as a function of their characteristics with a model, and it uses the model to evaluate counterfactual outcomes from different policies. This type of analysis can help give decision-makers guidance about whether or not it is worthwhile to invest in further development and testing of an innovation that has not yet been deployed.

Iterating through the Cycle.—Once the analysis in Stage 4 completes, the cycle begins again at Stage 1, but now the researcher has access to new experimental data that becomes "historical" from the perspective of Stage 1. The cycle may continue many times as the product or service is improved. Over time, decisions such as the selection of outcome measures or the structure of the experiment may become more routine. However, some changes or modifications to interventions necessitate corresponding changes to the experimental design.

As described in Supplemental Appendix Section III, the Incremental Innovation Cycle can create both tactical and generalizable knowledge. For example, learning what works and for whom can create insight about mechanisms.

VIII. Conclusions

This paper presented a framework for analyzing the role of the economist as a designer in the innovation process for digital products in social impact applications. I identified six design roles for the economist: product designer, market designer, process designer, designer of empirical analyses, designer of outcome measures, and designer of complex experiments. I explained how the economist's toolkit was well suited to contribute to all of these roles. I also highlighted how tools from economics

and econometrics could be used to address challenges in design, including the six cross-cutting challenges emphasized in the paper. The ideas were illustrated with a number of case studies. I argued that digital technology is well suited to create direct social impact, while innovation in this area can create generalizable practical and scientific knowledge. I believe that the design and analysis of complex experiments has many open research questions and expect that it will prove to be fertile ground for some time to come, as innovation in digital products will continue to grow in importance in the economy.

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